

1. The Illinois Design Conference 2025 featured a firefighter providing a prompt to students at UIUC in order to make the tasks of firefighters easier. While my skills weren't used in IDC25, they were taken to Independence DSAI. With the help of Vedanth Iyer, we sketched up Rescue Redefined, which later became Raisq. It is meant to
 - Help firefighters detect victims during times of low-visibility, where lidar scanners would work better than a human eye.
 - Help a series of firefighters map out the building, as multiple Raisqf modules would work together to create a map of the building and help these first responders do the greatest good for the greatest number of people.
 - Monitor the firefighters' heart rate to alert other members of the team about possible safety concerns that may be associated with the first responder.
2. Since there aren't many datasets that are specifically meant for such a tool, we had to resort to tools that are meant for autonomous driving, as well as a dataset for mapping. We selected the following datasets
 - [nuScenes](#)
 - [NCLT](#)
 - [WAYMO Open](#)
 - [TerraMobilita/iQmulus](#)
 - [Argoverse 2](#)
3. The Google Colab can be located in the Github repository.
4. Not every dataset contains tabular data. And for the ones that do, none of them have 10 features to begin with. Feature Selection on our current datasets would be difficult, as our datasets are focused more to align with Convolutional Neural Networks. Additionally, merging the already existing features would be impossible, as each dataset is used to specialize in a certain area of the residue module.
5. It is true that data is rarely perfect. However, the way our data is formatted makes it incredibly difficult to adjust. Each dataset specializes in its own field, and may not even be tabular to begin with. As we cater to fire fighters, Convolutional Neural Networks paired with LiDAR devices would pose the best solution, and can easily be executed without tabular data.